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Mansoor Ahmed

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Education

Georgia State University / Georgia Tech (Joint) Aug 2024 – May 2028

- PhD Candidate in Computer Science. GPA: 4.0/4.0.

University of Engineering and Technology, Lahore, B.Sc. Electrical Engineering Dec 2017 – Aug 2021

- Full scholarship recipient by Higher Education Commission Pakistan

Research Experience

Algorithmic Biology Lab, Georgia State University — Graduate Research Assistant Aug 2024 – present

- Advised by **Prof. Murray Patterson** & **Prof. Hemanth Venkateswara**.
- Developing theoretical generative modeling frameworks with applications to antibody design.
- First-author papers published at TMLR, and ICLR, ICML, and NeurIPS workshops (see below).

Lab for Understanding Network Effects (LUNE), UCLA — Research Fellow Aug 2025 – May 2026

- Advised by **Prof. Jennifer Wilson**. Drug combination modeling for cancer subtypes via network methods.

Data Analytics Lab, LUMS — Research Assistant (Lahore, Pakistan) Dec 2022 – June 2024

- Advised by **Prof. Imdadullah Khan**. Graph-based methods for antigen-antibody interactions.

Smart Data Systems Lab, LUMS — Research Assistant (Lahore, Pakistan) Nov 2021 – Nov 2022

- Advised by **Prof. Zubair Khalid**. Software development and database management for IoT devices.

Teaching Experience

Advanced Machine Learning (CSC 8850) — Georgia State University Spring 2025; Fall 2025, 2026

- Graduate Teaching Assistant with **Prof. Sergey Plis**. Held office hours; created and graded problem sets and exams; delivered lectures on manifold learning.

Computer Org. & Programming (CS 3210) — Georgia State University Summer 2025, Spring 2026

- Teaching Assistant. Assembly language and computer architecture.

Spatial and Scientific Databases (CSC 8713) — Georgia State University Fall 2024

Mathematical Foundations for Machine Learning (EE 212) — LUMS Fall 2022

Honors & Awards

Molecular Basis for Disease (MBD) Fellowship, Georgia State University 2025 – present

Warren Alpert Computational Biology and AI Network Fellow, UCLA 2025 – 2026

Computational Genomics Summer Institute (CGSI), UCLA 2025

EMBO Latin American Workshop on AI for Protein Design, Chile 2025

International Summer School in Biomedical Engineering, Max Planck Institute 2023

Higher Education Commission Undergraduate Scholarship, Pakistan 2017 – 2021

Publications

Accepted Publications

1. **ChiMERa-Bench: A Benchmark Dataset for Epitope-specific Antibody Design** 2026
   M. Ahmed, N. Taj, I. U. Khan, H. Venkateswara, **M. Patterson**.
   Transactions on ML Research (TMLR); ICLR GEM Workshop; 42+ GitHub stars

2. **Latent-Kernel Discrete Diffusion for Few-Step Generation** 2027
   M. Ahmed, Y.T. Fan, H. Venkateswara, **M. Patterson**. *NeurIPS BeNTo Workshop.*

3. **EvoStruct: Bridging Evolutionary and Structural Priors for Antibody CDR Design via Protein Language Model Adaptation** 2026
   M. Ahmed, S. Lee, U. Khayaz, **M. Patterson**. *ICML GenBio Workshop.*

4. **ConTact: Contact-First Antibody CDR Design via Explicit Interface Reasoning** 2026
   M. Ahmed, S. VonBank, Nadeem Taj, S. Lee, N. Jan, **M. Patterson.** *ICML GenBio Workshop.*

5. **Improved Graph-based Antibody-aware Epitope Prediction with PLM Embeddings** 2025
   M. Ahmed, S. Ali, A. Jan, I. U. Khan, **M. Patterson**. *ICCABS. Springer LNBI post-proceedings.*

6. **Robust Brain Age Estimation via Regression Models and MRI-derived Features** 2023
   M. Ahmed, U. Sardar, S. Ali, S. Alam, **M. Patterson**, **I. U. Khan**. *ICCCI. Springer CCIS.*

Preprints & Under Review

1. **MLD3: Mixture-of-Latents for Few-Step Discrete Diffusion Distillation** 2027
   M. Ahmed, S. Tang, H. Venkateswara, **M. Patterson**. *Under review at ICLR.*

2. **Domino: Joint Unmasking via First-order Markov Chain for Few-Step Masked Diffusion** 2026
   M. Ahmed, H. Venkateswara, **M. Patterson**. *Under review at TMLR.*

3. **EpiFormer: Learning Antigen-Antibody Interactions for Epitope Prediction** 2027
   M. Ahmed, H. Chai, H. Wang, H. Venkateswara, **M. Patterson**. *Under review at ICLR.*

4. **AgForce Enables Antigen-conditioned Generative Antibody Design** 2026
   M. Ahmed, **M. Patterson**. *Under review at AAAI (phase 2).*

Reviewing & Service

Workshops: GenBio @ ICML 2026, GEM @ NeurIPS 2026, ML4H @ NeurIPS 2026, BeNTo @ NeurIPS 2026

Conferences: AAAI 2027, ICLR 2027, ISBRA 2025; 2026, ICCABS 2025

Journals: PeerJ Computer Science, TMLR

Skills

ML/DL: PyTorch, PyTorch Geometric, e3nn, PyTorch Lightning, Hugging Face, WandB

Protein ML: ESM-2, AntiBERTy, ProteinMPNN, AlphaFold2, OpenFold, BoltzGen, RFdiffusion

Structural Biology: BioPython, PyMOL, RosettaDock, AutoDock, BLAST, ANARCI

Programming: Python, C++, CUDA, SQL, Git, Unix/Shell, LaTeX, SLURM, Docker